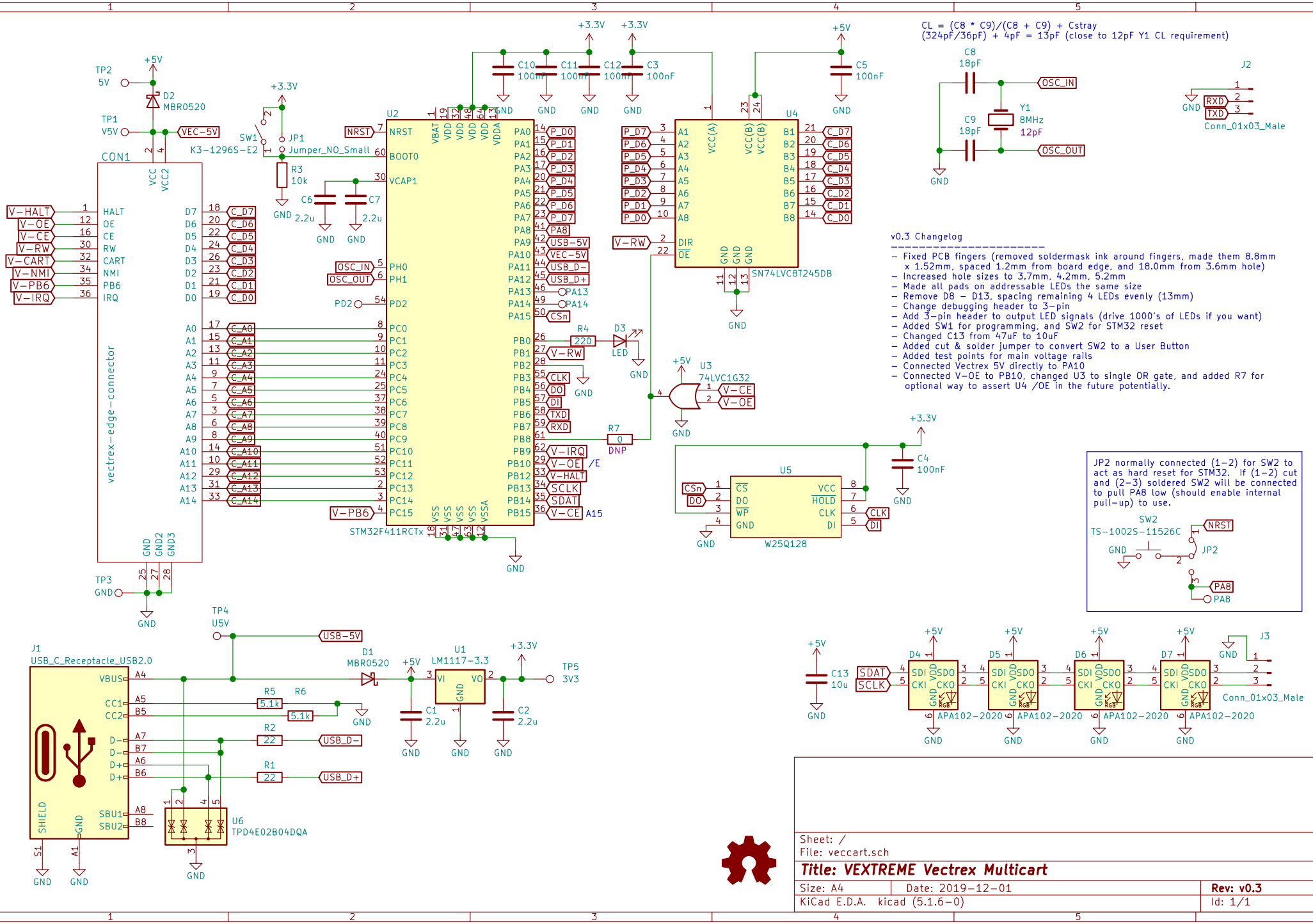

CL = (C8 * C9)/(C8 + C9) + Cstray
(324pF/36pF) + 4pF = 13pF (close to 12pF Y1 CL requirement)
v0.3 Changelog
– Fixed PCB fingers (removed soldermask ink around fingers, made them 8.8mm x 1.52mm, spaced 1.2mm from board edge, and 18.0mm from 3.6mm hole)
– Increased hole sizes to 3.7mm, 4.2mm, 5.2mm
– Made all pads on addressable LEDs the same size
– Remove D8 – D13, spacing remaining 4 LEDs evenly (13mm)
– Change debugging header to 3-pin
– Add 3-pin header to output LED signals (drive 1000's of LEDs if you want)
– Added SW1 for programming, and SW2 for STM32 reset
– Changed C13 from 47uF to 10uF
– Added cut & solder jumper to convert SW2 to a User Button
– Added test points for main voltage rails
– Connected Vectrex 5V directly to PA10
– Connected V–OE to PB10, changed U3 to single OR gate, and added R7 for optional way to assert U4 /OE in the future potentially.
JP2 normally connected (1–2) for SW2 to act as hard reset for STM32. If (1–2) cut and (2–3) soldered SW2 will be connected to pull PA8 low (should enable internal pull–up) to use.
CON1
vectrex–edge–connector
U2
STM32F411RCTx
U4
SN74LVC8T245DB
U5
W25Q128
U3
74LVC1G32
U1
LM1117–3.3
U6
TPD4E02B04DQA
J1
USB_C_Receptacle_USB2.0
J2
Conn_01x03_Male
J3
Conn_01x03_Male
SW1
K3–1296S–E2
SW2
TS–1002S–11526C
D1
MBR0520
D2
MBR0520
D3
LED
D4
APA102–2020
D5
APA102–2020
D6
APA102–2020
D7
APA102–2020
Y1
8MHz
12pF
Sheet: /
File: veccart.sch
Title: VEXTREME Vectrex Multicart
Size: A4
Date: 2019–12–01
Rev: v0.3
KiCad E.D.A. kicad (5.1.6–0)
Id: 1/1